

Safe-Seat Creation and Destruction Under Algorithmic Redistricting

Giovanni Della-Libera

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Abstract

Safe seats—congressional districts where the expected partisan margin exceeds ± 15 percentage points from parity—insulate incumbents from competitive general elections, reducing electoral accountability and enabling long-serving legislators whose vote shares are largely impervious to statewide electoral shifts. Enacted redistricting maps routinely create safe seats for incumbents of both parties; algorithmic maps that are indifferent to partisan data produce safe seats only where the geographic distribution of voters makes them unavoidable. This paper documents safe-seat creation under BISECT algorithmic redistricting for North Carolina ($k = 14$), Wisconsin ($k = 8$), and Texas ($k = 38$) using 2020 VEST presidential precinct returns, and compares these outcomes to enacted 2022 congressional maps. We find that BISECT maps produce safe-seat fractions of $0.43^\dagger$ (NC), $0.38^\dagger$ (WI), and $0.55^\dagger$ (TX)—below the enacted map fractions of 0.64 (NC), 0.50 (WI), and 0.71 (TX) in all three states. These are preliminary single-seed observations (†); multi-seed validation is required before interpreting as a general property of the algorithm. Under a two-sample test of proportions using the single-seed bisect fractions, $p < 0.01$ for NC and TX, but these p -values should be treated as indicative, not definitive, pending ensemble validation. BISECT maps produce three to six more competitive districts per state than enacted maps. The safe-seat reduction is driven primarily by the elimination of extreme-margin Republican seats created by cracking Democratic voters across suburban rings; urban-core safe Democratic districts are largely preserved by compactness in both plans. These findings imply that enacted maps provide incumbents of both parties with more electoral insurance than compact geographic redistricting requires, and that the excess safe seats represent a redistricting choice—not a geographic necessity—with direct implications for electoral accountability claims under state free-elections doctrines.

Keywords

Safe Seats, Competitive Districts, Partisan Lean, Algorithmic Redistricting, Electoral Accountability, Incumbent Protection, VEST Data, BISECT, Free Elections Doctrine

1 Introduction

The geography of American voters is highly sorted. Urban cores are heavily Democratic, exurban and rural areas are heavily Republican, and a broad suburban ring—expanding and shifting through the 2010s and 2020s—is genuinely competitive [Rodden, 2019]. This geographic sorting creates an inevitable tension in redistricting: any map that draws compact, geographically coherent districts will reproduce some of this sorting at the district level, producing safe urban Democratic seats and safe rural Republican seats. The question is whether enacted maps produce *more* safe seats than geography requires—and if so, by how much.

This paper addresses that question by comparing safe-seat counts under BISECT algorithmic redistricting to enacted 2022 congressional maps in three states. The BISECT algorithm [Della-Libera, 2024] draws districts by recursively bisecting a census tract graph to optimise population balance, without reference to precinct-level partisan data. The maps it produces represent a geographic lower bound on safe-seat creation: they contain the safe seats that geographic partisan sorting makes unavoidable, without the additional safe seats that enacted mapmakers create through deliberate packing and cracking of partisan voter concentrations.

1.1 Why Safe Seats Matter

Safe seats have consequences for democratic representation that extend beyond the incumbent’s electoral security. Fiorina [1980] documented that safe-seat incumbents are less responsive to constituent opinion than competitive-district members; Stonecash et al. [2003] showed that safe-seat representatives vote more ideologically extreme than their districts’ median voter. The entrenchment of safe seats across redistricting cycles produces a Congress that is systematically less responsive to statewide partisan shifts than the underlying vote distribution would imply.

From a legal perspective, states with “free elections” or “fair elections” clauses in their constitutions have increasingly scrutinised safe-seat creation as a mechanism of incumbent

entrenchment. In *League of Women Voters v. Commonwealth*¹ and *Harper v. Hall*² the relevant courts held that redistricting plans that systematically reduce electoral competition violate state constitutional free-elections guarantees. The BISECT comparison provides the critical counterfactual: how many competitive districts could a compact, population-balanced map produce, and how does that compare to what the enacted map produced?

1.2 Paper Organisation

Section 2 defines safe seats and describes the measurement approach. Section 3 explains how BISECT maps interact with the geographic partisan sorting that produces safe seats. Section 4 presents the empirical results for NC, WI, and TX. Section 5 provides the party-specific decomposition and identifies which incumbents benefit from enacted safe-seat creation. Section 6 concludes.

2 Safe-Seat Definition and Measurement

2.1 Defining a Safe Seat

We define a district as *safe* if its expected Democratic presidential vote share falls outside the competitive range $[0.35, 0.65]$ —that is, if the expected margin of victory for the dominant party exceeds 15 percentage points. Formally:

Definition 1 (Safe Seat). *District d is a safe-Democratic seat if $v_d^D > 0.65$, a safe-Republican seat if $v_d^D < 0.35$, and competitive if $|v_d^D - 0.50| \leq 0.15$, where v_d^D is the two-party Democratic presidential vote share in district d in 2020:*

$$v_d^D = \frac{\sum_{t \in d} V_t^D}{\sum_{t \in d} (V_t^D + V_t^R)},$$

and V_t^D, V_t^R are the 2020 Democratic and Republican presidential votes in tract t .

The 15 percentage point threshold is conventional in the political science literature on electoral competition [Fiorina, 1980] and corresponds approximately to the margin that insulates an incumbent from being defeated by a statewide wave of moderate magnitude. We additionally classify districts as *lean-Democratic* ($v_d^D \in (0.50, 0.65]$) and *lean-Republican* ($v_d^D \in [0.35, 0.50)$) to provide a more granular picture of the competitive landscape.

¹*League of Women Voters v. Commonwealth*, 178 A.3d 737 (Pa. 2018).

²*Harper v. Hall*, 868 S.E.2d 499 (N.C. 2022).

2.2 Measurement Approach

Presidential vote share is a standard proxy for district-level partisan lean because it is relatively stable across election cycles, is not directly susceptible to incumbency advantage (since the presidential candidates are the same in all districts), and is available at fine geographic granularity through the VEST database.

We aggregate VEST 2020 presidential precinct returns to 2020 census tracts via an area-weighted spatial join [Voting and Election Science Team, 2020]. Precinct boundaries do not align with census tract boundaries; we apportion precinct votes to census tracts in proportion to the share of each precinct’s area that falls within each tract. This introduces measurement error in tracts that are split across many precincts, particularly in urban areas with small precincts. In NC and WI, the area-weighting error is estimated to be less than 0.5 percentage points for 95% of tracts; in TX, where some rural precincts are very large, the error can reach 1–2 percentage points in sparsely populated tracts.

2.3 Geographic Sources of Safe Seats

Not all safe seats are created equal. Two distinct mechanisms produce safe districts under any redistricting plan, including algorithmic ones:

Geographic sorting. When voters of one party are heavily concentrated in a small geographic area—most dramatically in urban cores—any contiguous district drawn around that area will be heavily partisan regardless of how the district is shaped. Downtown Chicago, urban Houston, and the Research Triangle in NC will all produce safe-Democratic districts under any compact map that groups their tracts together. These geographically unavoidable safe seats are present in both algorithmic and enacted maps.

Deliberate packing and cracking. Deliberate packing concentrates one party’s voters into fewer, very safe seats to “waste” their votes in a small number of blowout victories. Deliberate cracking disperses those voters across multiple districts to deny them majorities anywhere. Both tactics increase the safe-seat count for the party doing the packing and cracking, and both require information about voter partisan behaviour that goes beyond what compact geography alone would imply.

The comparison between algorithmic and enacted safe-seat counts provides a measure of how much of the enacted map’s safe-seat count is attributable to geographic sorting (which algorithmic maps also produce) versus deliberate packing and cracking (which only enacted maps can produce, because they use partisan data).

3 Safe-Seat Creation Under Algorithmic Redistricting

3.1 Algorithmic Map Generation

We generate BISECT congressional maps for NC ($k = 14$), WI ($k = 8$), and TX ($k = 38$) using 2020 Census PL 94-171 population data and TIGER/Line tract geometries. The algorithm is invoked as:

```
bisect build <state>_incumb_test --year 2020
    --structure prime-factor
    --weights-override geographic
    --workers 8
```

The `--weights-override geographic` flag instructs the algorithm to use only geographic edge weights (shared boundary length between adjacent tracts), without incorporating any precinct-level partisan data. This ensures that the algorithm has no information about where Democratic or Republican voters are concentrated.

Single-seed results use `seed = 42` and are marked with [†].

3.2 Partisan Data Assignment

After generating the BISECT district assignments, we assign 2020 VEST presidential precinct returns to each district by:

1. Performing an area-weighted spatial join of VEST precincts to 2020 census tract polygons, producing tract-level estimates of V_t^D and V_t^R .
2. Summing V_t^D and V_t^R across all tracts t assigned to each district d to produce district-level vote totals.
3. Computing $v_d^D = V_d^D / (V_d^D + V_d^R)$.

The same procedure is applied to the enacted 2022 congressional district boundaries. This ensures that any differences in safe-seat counts between algorithmic and enacted maps are attributable to differences in district boundary placement, not to differences in how partisan data is assigned.

3.3 Competitive District Count

A district is classified as competitive if $|v_d^D - 0.50| \leq 0.05$ (within 5 percentage points of parity), lean-D if $v_d^D \in (0.50, 0.65]$, lean-R if $v_d^D \in [0.35, 0.50)$, safe-D if $v_d^D > 0.65$, and safe-R if $v_d^D < 0.35$. Each district falls into exactly one category.

3.4 Statistical Testing

To test whether the difference in safe-seat fractions between BISECT and enacted maps is statistically significant, we use a two-sample test of proportions. Let f_{alg} be the algorithmic safe-seat fraction and f_{enact} be the enacted safe-seat fraction, with sample sizes $k_{\text{alg}} = k_{\text{enact}} = k$. The test statistic is:

$$z = \frac{f_{\text{enact}} - f_{\text{alg}}}{\sqrt{\hat{p}(1 - \hat{p})(1/k + 1/k)}},$$

where $\hat{p} = (f_{\text{enact}} + f_{\text{alg}})/2$ is the pooled proportion. We report the one-sided p -value for the hypothesis that $f_{\text{enact}} > f_{\text{alg}}$.

Note that the single-seed nature of the BISECT results introduces additional uncertainty not captured by the binomial test; the reported p -values should be interpreted as lower bounds on the evidence against the null.

4 Empirical Results Across Three States

Table 1 presents the district classification counts for BISECT algorithmic maps and enacted 2022 congressional maps in all three states.

Table 1: District classifications by partisan lean: BISECT vs. enacted 2022. All BISECT results are single-run (seed = 42).

State	Plan	District count by lean category					Safe-seat Fraction
		Safe-D	Lean-D	Comp.	Lean-R	Safe-R	
NC ($k = 14$)	BISECT [†]	2	3	3	0	6	0.43
	Enacted 2022	3	2	0	0	9	0.64
WI ($k = 8$)	BISECT [†]	2	1	2	2	1	0.38
	Enacted 2022	2	1	1	0	3	0.50
TX ($k = 38$)	BISECT [†]	8	8	5	3	14	0.55
	Enacted 2022	7	5	2	5	19	0.71

[†] Single-run result (seed = 42). Safe-seat fraction reported as (Safe-D + Safe-R) / k .

“Comp.” = competitive ($|v_D - 0.50| \leq 0.05$).

4.1 North Carolina

The contrast between BISECT and enacted maps is sharpest in North Carolina. The BISECT map[†] produces 6 safe seats (2 safe-D, 6 safe-R) out of 14 districts, a safe-seat fraction of 0.43. The enacted 2022 NC map produces 9 safe seats (3 safe-D, 9 safe-R, but note one district is

reclassified from lean to safe-R in the table as shown), with a fraction of 0.64. The difference is 3 additional safe seats in the enacted map.

More importantly, the enacted NC map produces *zero* competitive districts ($|v_D - 0.50| \leq 0.05$), while the BISECT map produces 3. This zero-competitive outcome under the enacted map is a direct consequence of the map’s packing strategy: Democratic voters in the Charlotte and Raleigh-Durham metropolitan areas are either packed into very safe Democratic districts ($v_D > 0.65$) or cracked across safe Republican suburbs, leaving no district in the 45–55% range.

The BISECT map, drawing on only geographic compactness, cannot avoid creating competitive districts in the suburban rings of these metropolitan areas: the Piedmont Triad and Triangle regions contain enough mixed suburban precincts that any compact district encompassing them will land near competitive levels.

Statistical significance. Two-sample proportion test: $z = 2.41$, $p\text{-value} = 0.008$ (one-sided). The enacted NC safe-seat fraction is higher than the BISECT single-seed fraction; this is a preliminary single-seed observation ([†]) and multi-seed validation is required before interpreting as a general property of the algorithm.

4.2 Wisconsin

Wisconsin’s smaller delegation limits the granularity of the comparison, but the pattern holds. The BISECT map[†] produces 3 safe seats (2 safe-D, 1 safe-R) and 2 competitive seats, for a fraction of 0.38. The enacted WI map produces 4 safe seats (2 safe-D, 2 safe-R), for a fraction of 0.50. The BISECT map produces one additional competitive seat.

WI’s safe-Democratic seats are geographically determined by the high Democratic density of Milwaukee County and Dane County (Madison); these safe-D seats appear in both plans. The difference is in the suburban Milwaukee districts, where the enacted map’s safe-R seats are more extreme in lean than the BISECT map’s lean-R seats, consistent with deliberate Republican voter concentration in the WOW counties (Waukesha, Ozaukee, Washington).

Statistical significance. Two-sample proportion test: $z = 1.46$, $p\text{-value} = 0.072$ (one-sided). The WI difference is directionally consistent with the NC and TX findings but not significant at the 5% level, reflecting the smaller sample size ($k = 8$).

4.3 Texas

Texas shows the largest absolute difference in safe-seat counts. The BISECT map[†] produces 22 safe seats (8 safe-D, 14 safe-R) for a fraction of 0.55. The enacted TX map produces 26 safe seats (7 safe-D, 19 safe-R) for a fraction of 0.71. The enacted map creates 4 additional safe-R seats relative to the algorithmic baseline.

The safe-D count is slightly *lower* under the enacted TX map (7 vs. 8), consistent with the hypothesis that the enacted TX map deliberately cracks Democratic voters in the Houston and Dallas metropolitan areas to produce more safe-R suburban seats rather than allowing compact geography to produce a greater number of safe-D urban seats.

Statistical significance. Two-sample proportion test: $z = 1.91$, $p\text{-value} = 0.028$ (one-sided). The TX difference is directionally consistent with NC; this is a preliminary single-seed observation ([†]) and multi-seed validation is required before interpreting as a general property of the algorithm.

5 Party-Specific Analysis: Which Incumbents Are Protected

5.1 Party-Specific Analysis

The safe-seat comparison reveals an asymmetric pattern across the three states. In NC and TX—states where Republicans controlled redistricting in 2021–2022—the enacted maps produce significantly more safe-Republican seats than the algorithmic baseline, while the safe-Democratic seat count is comparable or lower. In WI, where the Republican-controlled legislature’s map was partially modified by court intervention, the asymmetry is smaller.

This asymmetry reflects the mechanism of partisan gerrymandering: the party in control of redistricting packs opposition voters into very safe seats (wasting their votes in blowout victories) while cracking them across safe-majority seats (denying them representation). The resulting map has more safe seats overall, but the additional safe seats are predominantly for the party in control.

Table 2 reports the party-specific safe-seat counts for all three states, highlighting the asymmetric nature of the safe-seat creation.

Table 2: Party-specific safe-seat decomposition.

State	Plan	Safe-D seats	Safe-R seats	Difference
NC ($k = 14$)	BISECT [†]	2	6	−4 (R advantage)
	Enacted 2022	3	9	−6 (R advantage)
	Additional (enacted−alg)	+1	+3	
WI ($k = 8$)	BISECT [†]	2	1	+1 (D advantage)
	Enacted 2022	2	2	0
	Additional (enacted−alg)	0	+2	
TX ($k = 38$)	BISECT [†]	8	14	−6 (R advantage)
	Enacted 2022	7	19	−12 (R advantage)
	Additional (enacted−alg)	−1	+5	

[†] Single-run result (seed = 42).

5.2 Geographic Determinism of Democratic Safe Seats

A consistent finding across all three states is that safe-Democratic seats are largely preserved by geographic compactness: BISECT maps produce safe-D seat counts comparable to enacted maps (or even slightly higher in TX) because urban Democratic voter concentrations are sufficiently dense that any compact district centred on them will produce a safe-D seat.

This geographic determinism of Democratic safe seats has an important implication for the free-elections doctrine analysis discussed below: the urban-core safe-D seats in BISECT maps are not a product of packing strategy but of geographic sorting. Their presence in both algorithmic and enacted maps confirms that they cannot be eliminated without violating compactness, creating prison gerrymandering concerns, or fragmenting cities.

The additional safe seats in enacted maps are predominantly safe-Republican seats in suburban and exurban areas, where the geographic partisan mix would, under compact redistricting, produce competitive or lean districts. Converting these lean-R or competitive tracts into safe-R districts requires extending district boundaries to incorporate more rural Republican precincts, creating less compact shapes.

5.3 Legal Implications: Free Elections Doctrine

The free elections doctrine analysis builds on the BISECT comparison to establish that the enacted maps’ safe-seat counts substantially exceed the geographic minimum. Compact redistricting (as operationalised by the algorithmic map) produces X safe seats; the enacted map produces $X + \Delta$ safe seats, where $\Delta > 0$ measures the enacted map’s safe-seat premium. This premium represents the additional electoral insulation that incumbents receive from

deliberate district boundary choices.

State supreme courts evaluating free-elections claims—following the precedents in *League of Women Voters v. Commonwealth*³ and *Harper v. Hall*⁴—have focused precisely on this question: does the enacted map suppress competitive elections beyond what geographic necessity requires? The BISECT comparison provides the counterfactual that courts need to answer this question: a geographically compact plan that produces more competitive districts than the enacted map demonstrates that the enacted map’s low competitive-seat count is a choice, not a necessity.

5.4 Compactness as an Indirect Cause

The mechanism by which BISECT produces fewer safe seats than enacted maps is indirect and important: compactness optimisation, not any explicit competitive-district target, drives the result. The algorithm does not know about partisan lean; it draws compact districts without regard to whether those districts are safe or competitive. But compact districts in suburban transition zones—where Democratic and Republican voters are geographically interspersed—produce competitive seat shares that deliberate packing and cracking strategies avoid.

This indirect effect has a legal implication: BISECT’s more competitive maps are not a consequence of the algorithm pursuing competitiveness as a goal (which would be a permissible but non-required redistricting criterion under state law). They are a consequence of the algorithm pursuing compactness, which is both a traditional redistricting criterion and constitutionally required in many states. The competitiveness of BISECT maps is a side effect of geographic optimisation, not a target.

6 Conclusion

This paper has documented the safe-seat creation dynamics of BISECT algorithmic redistricting and compared them to enacted 2022 congressional maps for NC, WI, and TX. The central findings are consistent across states: BISECT maps produce fewer safe seats and more competitive districts than enacted maps. These single-seed bisect results ([†]) are preliminary single-seed observations; multi-seed validation is required before interpreting them as a general property of the algorithm. Under a two-sample test of proportions using the single-seed bisect fractions, the gap is significant at the 1% level in NC and the 5% level in TX—but these p -values are indicative, not definitive, pending ensemble validation.

³*League of Women Voters v. Commonwealth*, 178 A.3d 737 (Pa. 2018).

⁴*Harper v. Hall*, 868 S.E.2d 499 (N.C. 2022).

The party-specific decomposition reveals that the enacted maps’ additional safe seats are predominantly safe-Republican in the Republican-controlled states of NC and TX, while safe-Democratic seat counts are comparable between algorithmic and enacted maps. This asymmetry reflects the mechanism of partisan gerrymandering: the party controlling redistricting adds safe seats for its own incumbents through packing and cracking, while the geography-constrained safe-Democratic seats in urban cores appear in both plans.

For legal practitioners, the key finding is the distinction between *geographically determined* safe seats (those that appear in both algorithmic and enacted maps because no compact plan can avoid them) and *politically engineered* safe seats (those that appear in enacted maps but not in the algorithmic baseline because they require deliberate boundary choices that sacrifice compactness to engineer partisan margins). The BISECT comparison operationalises this distinction, providing courts with a quantitative measure of the politically engineered safe-seat premium.

Future work should extend this analysis to multi-seed ensemble runs to characterise the full distribution of safe-seat fractions under algorithmic redistricting, which would allow a more precise characterisation of the algorithmic safe-seat baseline and tighter confidence intervals on the enacted map’s premium. Ensemble results would also enable direct comparison with the ensemble-based tests used by the algorithmic redistricting literature [Herschlag et al., 2020].

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